Reg. No. | | | | | | | | | | | [13EC662/12EC662]

SIXTH SEMESTER B.TECH/B.E UNIVERSITY EXAMINATION, JUNE – 2016

DATA STRUCTURES USING C++

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. List the differences between Structure and Union. **05**

b. What is an array? How arrays can be declared in C/C++? Explain how one dimensional array can be implemented. **09**

2. a. What is a stack? What are the various operations that can be performed on stack? Explain with example. **09**

b. What is an expression? What are the various types of expressions and also mention any one application of stack. **05**

3. a. Convert the following infix expression to postfix expression. **06**

$$\left(\left(A+(B-C)*D\right)^{\wedge} E+F\right)$$

b. Obtain the postfix expression for the following **04**

$$\left(A+(B-C)*D\right)$$

c. Obtain the prefix expression for the following **04**

$$a=\hat{X}^{\wedge}\hat{Y}^{\wedge}Z^{\wedge}-M+N+P/Q$$

4. a. What is tower of Hanoi problem? Write a C/C++ function to implement this problem. **08**

b. What is Recursion? Explain about general case and Base case with example. **06**

5. a. What are the different types of Queues? Write a C/C++function to display the contents of Queue. **06**

b. Write a C program to create a list and to display the contents of list. **08**

6. a. Explain the following terminologies with respect to tree. **06**

i.	Root Node	**iv.**	Ancestors
ii.	Child	**v.**	Leaf
iii.	Siblings	**vi.**	Descendants

b. Write a C function for the following traversal techniques : **08**

i. Preorder
ii. In order
iii. Post order

7.	**a.**	Write a C/C++ program to arrange numbers in ascending order using bubble sort.	**08**
	b.	Sort the following elements using Radix sort 57, 45, 36, 91, 28, 79, 35, 68, 89, 20, 62, 43, 84, 55, 86, 96, 78, 67	**06**
8.	**a.**	Explain linear search with example.	**07**
	b.	List the differences between linear search and Binary Search and also mention the advantages and disadvantages of Binary search.	**07**